

RPI × CONTEXTUAL AI SYSTEMS

THE RPI AUTONOMOUS DRONE INITIATIVE

Applying autonomy to real-world safety, infrastructure and environmental challenges

- RCOS students build the first autonomous deterrence drone platform
- Real engineering
- Real modules
- Real ownership



Developing a system that protects wildlife, infrastructure and critical assets through autonomous systems



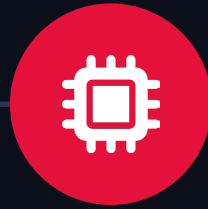
CONTEXTUAL·AI[™]
SYSTEMS

A MULTI-YEAR ENGINEERING PROGRAM



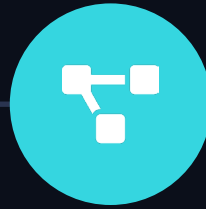
SEMESTER 1

Core autonomy modules



SEMESTER 2

Hardware integration + field tests



SEMESTER 3

Multi-drone coordination +
advanced AI



SEMESTER 4

Commercial prototypes + startup
formation



BEYOND

Grow with the project. Build
something lasting.

WHAT YOU WILL BUILD

Student-owned engineering tracks

- Computer Vision
- Motion Engine
- Navigation
- Safety Layer
- Docking
- Electronics: sensors, microcontrollers
- Simulation: full virtual test environment

“You pick a module. You own it. You ship real code.”



PATENTS & ENGINEERING HANDBOOK



Innovation Pipeline

- ✓ Autonomous deterrence drone
- ✓ Multi-sensor safety layer
- ✓ Autonomous docking system
- ✓ Wildlife deterrence algorithms
- ✓ Inspection autonomy



Open Engineering Platform

- ✓ Readme.md in Github
- ✓ Architectural diagram
- ✓ Module templates
- ✓ Simulation environment
- ✓ Good First Issues

RPI CAN OWN THIS SPACE



Creating a National Identity

MIT → Humanoid Robotics

CMU → Autonomous Ground Vehicles

Georgia Tech → Swarm Robotics

RPI → Autonomous Drones for Peaceful Use



RPI → Peaceful Autonomous Deterrence Drones

RPI has an opportunity to become the leading university for autonomous drone systems serving infrastructure, environmental and public-sector needs.

Questions?

Jeff Barat

jnbarat@dbnj.com

914.523.8950



Scan to learn more



WHO WE ARE SEEKING

Entrepreneurial students wanted

Builders, coders, tinkerers — module leads and team leads who want to ship something real.



Wanted

- Builders, coders, tinkerers
- Entrepreneurs
- Module leads
- Team leads



Requirements

- Curiosity
- Commitment
- Desire to build something real



Not Required

- Drone experience
- Robotics experience
- AI experience

Phase-0 MicroDrone

Build the autonomous microdrone that detects wildlife, makes decisions, and executes motion behaviors.

ARCHITECTURE/PHASE-0

CV → Navigation → Motion Engine → Safety Layer → Docking

WHAT YOU WILL BUILD THIS SEMESTER

- Synthetic Computer Vision pipeline
- State Vector + dead-reckoning
- Motor Command Stubs (no hardware)
- Safety Stubs (battery + emergency stop)
- Docking State Machine
- Simulation Environment

TECH STACK

Python · OpenCV · Jetson Nano · Gazebo/Webots · GitHub · VS Code



JOIN THE TEAM

Scan the QR code → Open the repo → Pick a module → Start contributing

Jeff Barat
jnbarat@dbnj.com
914-523-8950



CHOOSE YOUR ENGINEERING TEAM

- Computer Vision
- Motion Engine
- Navigation
- Safety Layer
- Docking

Questions?

Jeff Barat

jnbarat@dbnj.com

914.523.8950



Scan to learn more



CONTEXTUAL·AI™
SYSTEMS